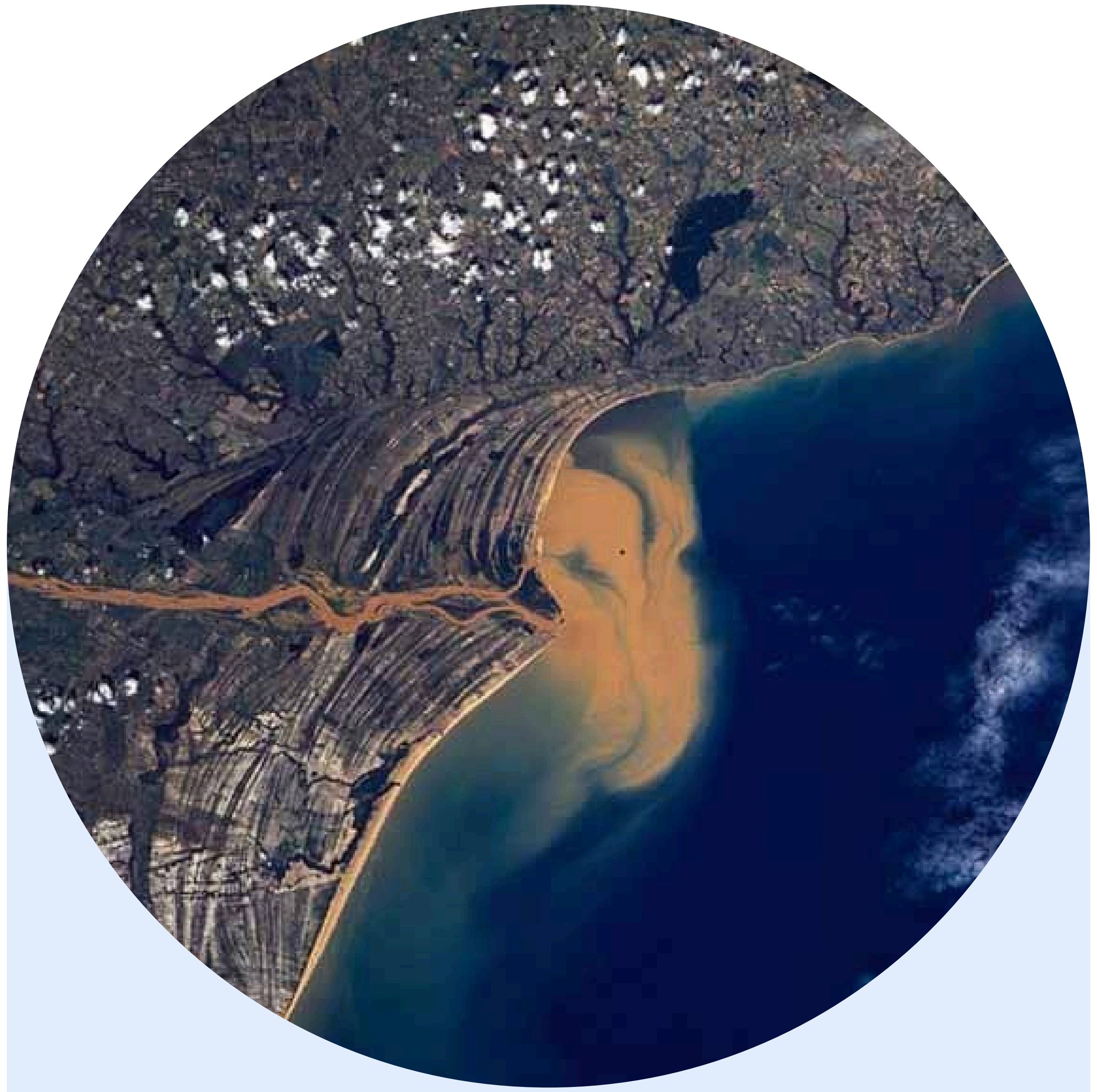
Field practices: Marine Monitoring

CASE STUDY: WATER CURRENT IMPACT ON THE CIVITANOVA COAST

Calocero Andrea, Savino Eleonora C., Sirianni Giulia, Sturiale Letizia B., Tessari Alessandro

Introduction



The relationship between humans and rivers has been in the past of fundamental importance for the progress of human cilvilization. However, continual anthropogenic pressure have led to a progressive deterioration of the water quality with rivers that can be sources of both natural and anthropogenic pollution.

The presence of river runoff can influence the achievement of the Good Environmental Status of sea water, according to the MSFD 56/2008

European Commission

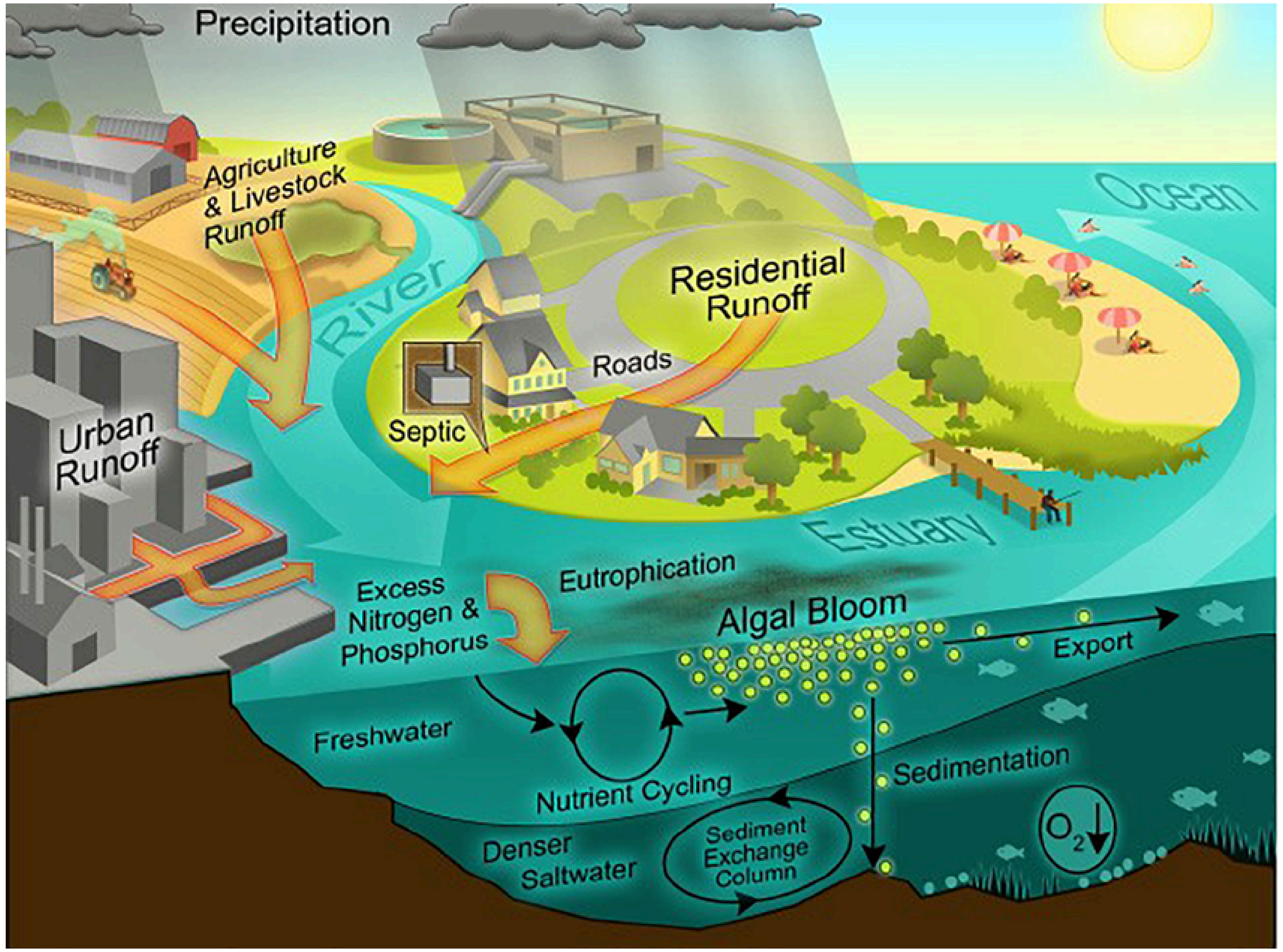
The Marine Strategy Framework Directive

Rivers impacts on marine ecosystems

- **INORGANIC NUTRIENTS INPUTS** coming from agriculture and urban runoff due to fertilizers, detergents, etc. (**D5**)

- **ORGANIC MATTER INPUTS** coming from the land system, called secondary organic matter (not of authotrophic origin) (**D5**)

- In case of **EUTROPHICATION** the area can be affected by **BIODIVERSITY LOSS,** mainly due to deoxygenation with alteration of the of the communities structure with replacement of sensitive species with more resistant, often opportunistic ones (**D5** - **D1**)



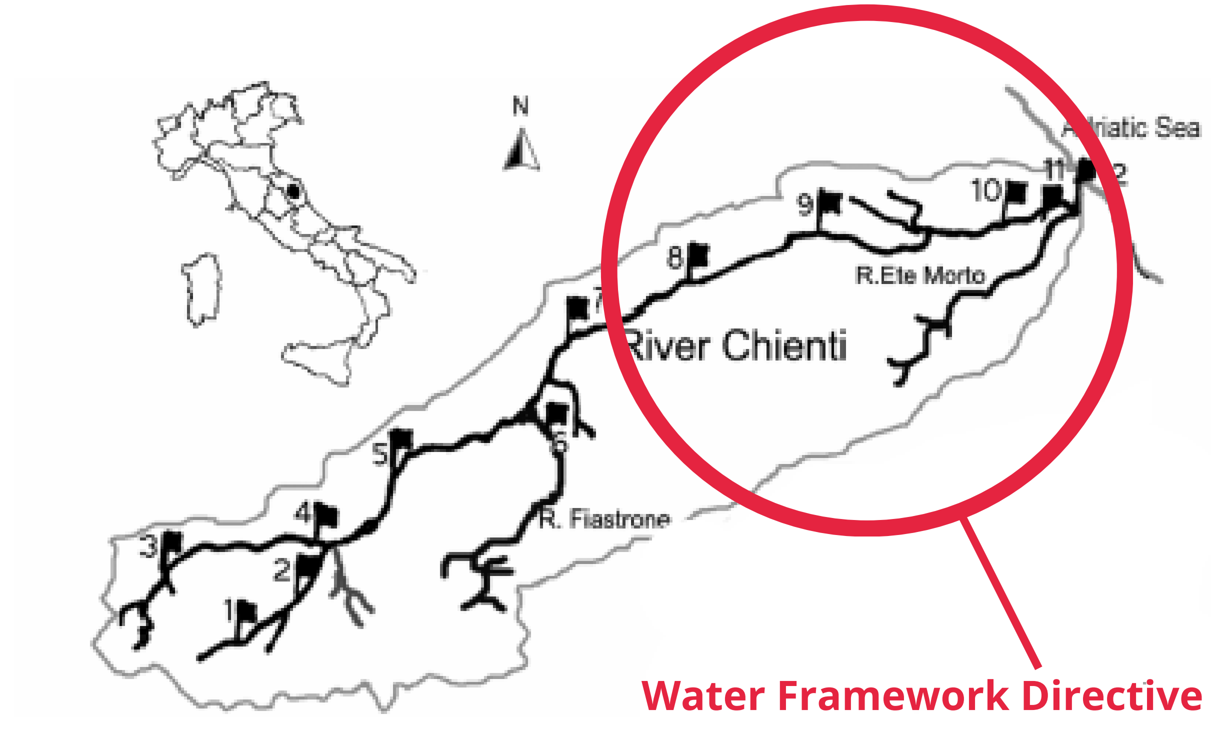
Malone, Thomas & Newton, Alice. (2020). The Globalization of Cultural Eutrophication in the Coastal Ocean: Causes and Consequences. Frontiers in Marine Science. 7. 10.3389/fmars.2020.00670.

CONTAINANTS INPUTS can be carried out by rivers, releasing into the marine ecosystem elements as plastics, microplastics and other contaminants, which can be directly discharged on them or washed by rain from the terrestrisl system, as hydrocarbons and heavy metals (**D8** - **D10**)

This contaminants can be **BIOACCUMULATED** in marine organisms and **BIO-MAGNIFICATED** along the trophic web with the possibility to reach humans too (**D9**)

Chienti Rivers

Chienti River is a watercourse of Central Appennines, it is 91 km long with maximum flow rate occurs in spring and minimun in late summer.



Water Framework Directive 2000/60/CE
Bad or Very Bad Quality

Scale1:400.000

(SCURI, Stefania, et al.)

The Western Adriatic Coastal Current (**WACC**) flows mainly **southward**. It transports sediments and nutrients from riverine sources. While **Bora** winds **strengthen** this southward flow, **Scirocco** winds can temporarily **slow down** it.

Courtney K. Harris,1 Christopher R. Sherwood, Richard P. Signell, Aaron J. Bever, and John C. Warner (2008). Sediment dispersal in the northwestern Adriatic Sea. JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 113, C11S03, doi: 10.1029/2006JC003868

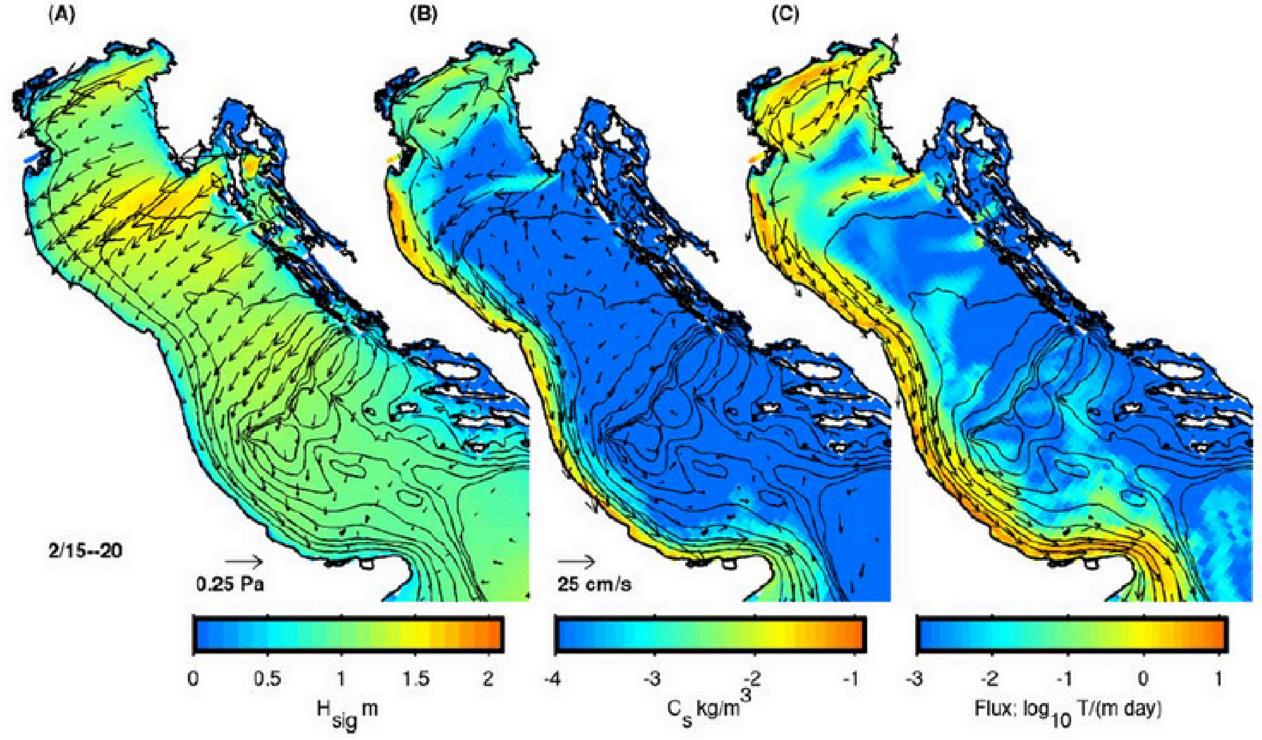


Figure 13. Conditions averaged during a Bora from 15 to 20 February 2003. (a) Wind stress (arrows; see scale for magnitude) and wave height (color). (b) Depth- and time-averaged sediment concentration (color) and current velocity (arrows; see scale for magnitude). (c) Time-averaged, depth-integrated flux (t m^{-1} d^{-1}). Arrows show direction where flux exceeds 0.1 t (m d)$^{-1}$. Contours drawn every 25 m up to 200-m water depth.

Hypotheses

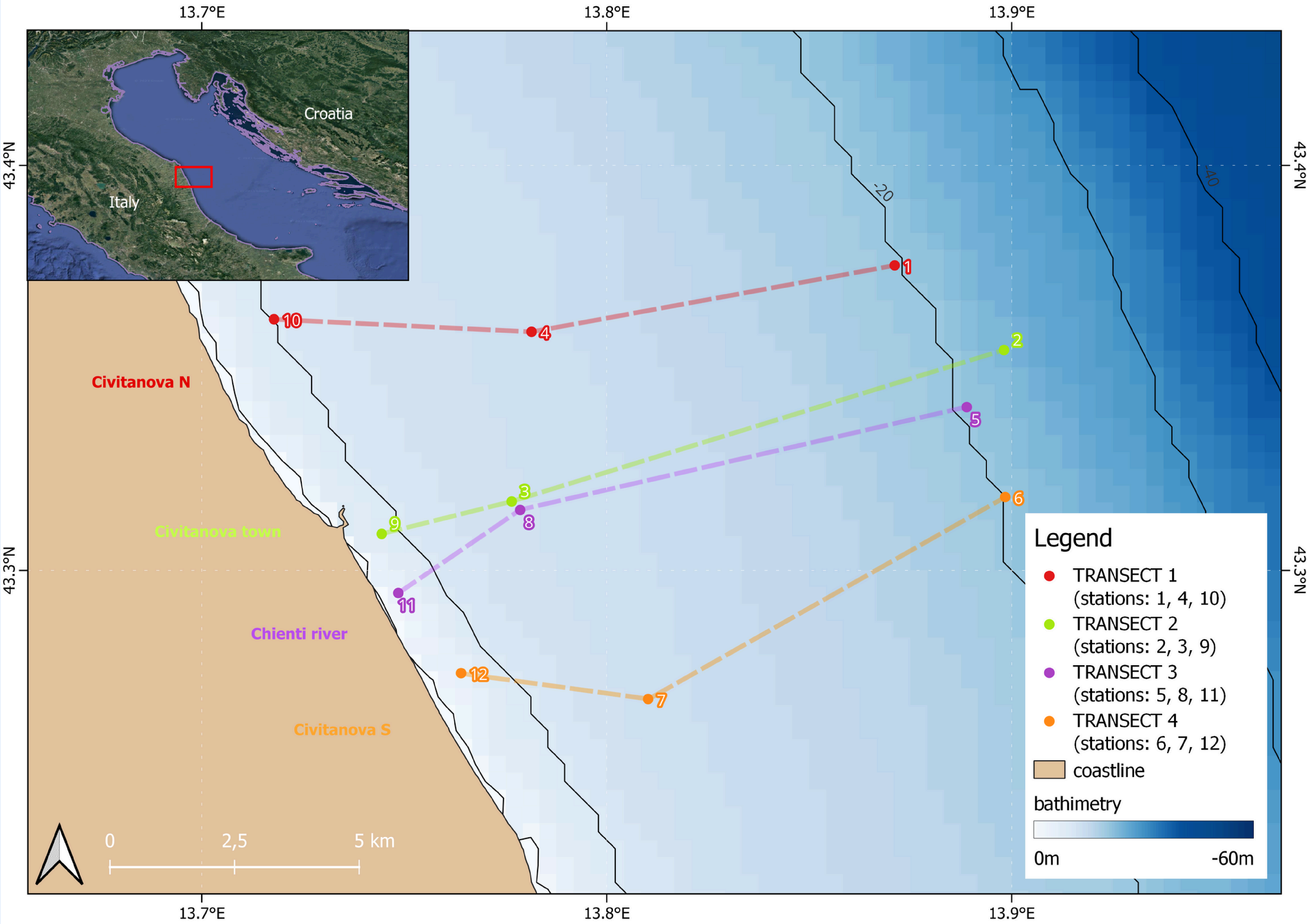
Ho

No significant impact of river runoff on environment or fauna; no current-driven pattern.

H1

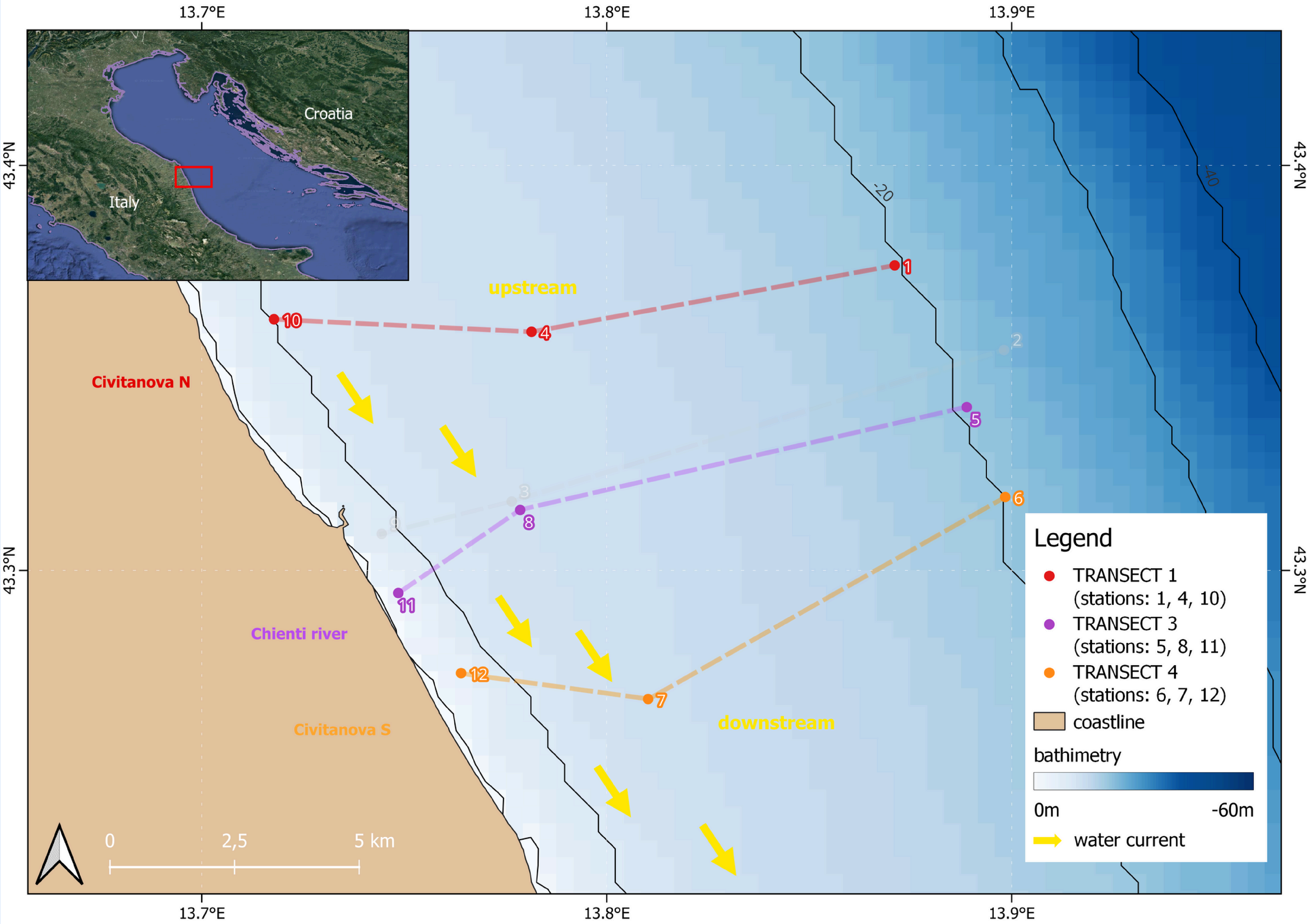
River runoff affects environmental quality and fauna, with stronger, asymmetrical impacts downstream due to coastal currents.

Study area



13.7°E
13.8°E
13.9°E
43.4°N
43.3°N
Croatia
Italy
-20
-40
1
2
3
4
5
6
7
8
9
10
11
12
Civitanova N
Civitanova town
Chienti river
Civitanova S
0
2,5
5 km
Legend
TRANSECT 1 (stations: 1, 4, 10)
TRANSECT 2 (stations: 2, 3, 9)
TRANSECT 3 (stations: 5, 8, 11)
TRANSECT 4 (stations: 6, 7, 12)
coastline
bathimetry
0m
-60m

Study area



13.7°E
13.8°E
13.9°E
43.4°N
43.3°N
Croatia
Italy
upstream
downstream
-20
-40
Civitanova N
Chienti river
Civitanova S
1
2
3
4
5
6
7
8
9
10
11
12
0
2,5
5 km
Legend
TRANSECT 1
(stations: 1, 4, 10)
TRANSECT 3
(stations: 5, 8, 11)
TRANSECT 4
(stations: 6, 7, 12)
coastline
bathymetry
0m
-60m
water current

Sampling activities on board

Research Vessel:
ANNA GUIDOTTI

- **IMO number:** 9096686
- **Type:** Offshore Supply Ship
- **Length (LOA):** 30.99 meters
- **Width (beam):** 6 meters

20th February 2025

Methodologies

Physical-Chemical Parameters

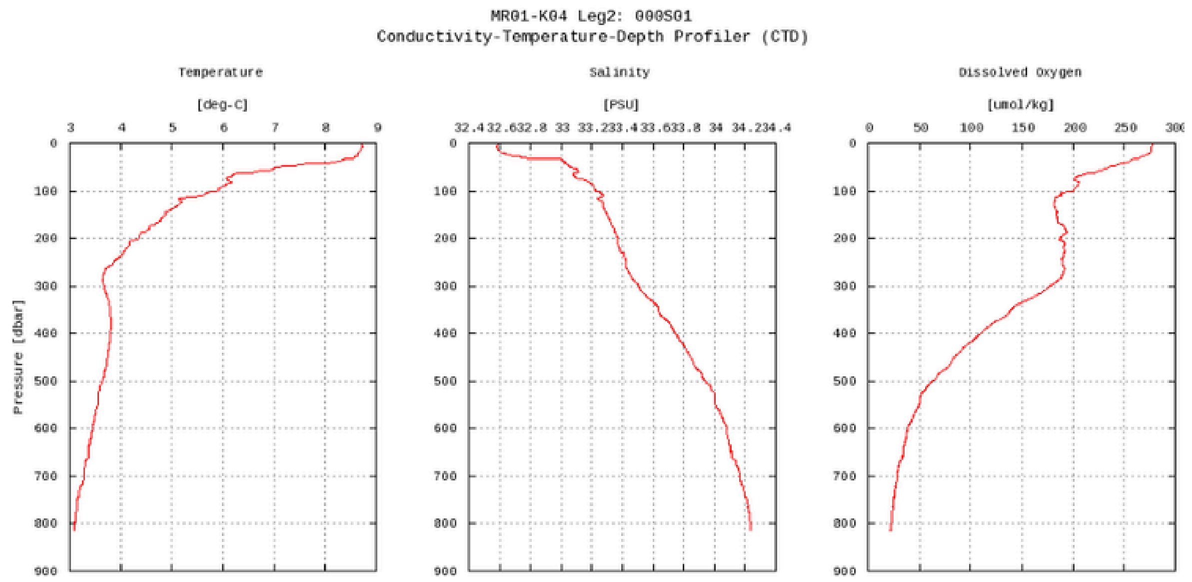
Biological Indicators
Water Column

Biological Indicators
Benthic Fauna

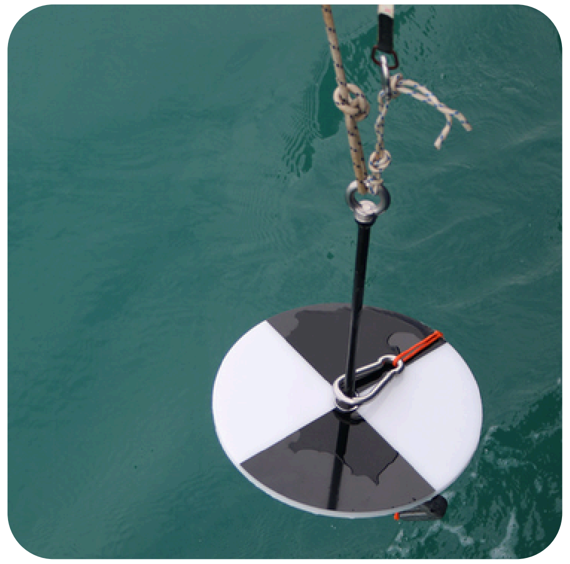
Methodologies

Physical-Chemical Parameters

1. CTD Profiling:
Measurement of Salinity, Temperature and Depth along the water column.



2. Turbidity:
Estimation of water transparency with Secchi Disk.



3. TRIX Index:
Trophic state index combining nutrients and Chl-a data.

Biological Indicators
Water Column

Biological Indicators
Benthic Fauna

Methodologies

Physical-Chemical Parameters

Biological Indicators
Water Column

1. Phytopigments (Chlorophyll-a): Indicator of phytoplankton biomass.

2. Zooplankton Analysis: Assessment of abundance, Shannon Index, and Pielou's Evenness.

Biological Indicators
Benthic Fauna

Methodologies

Physical-Chemical Parameters

Biological Indicators
Water Column

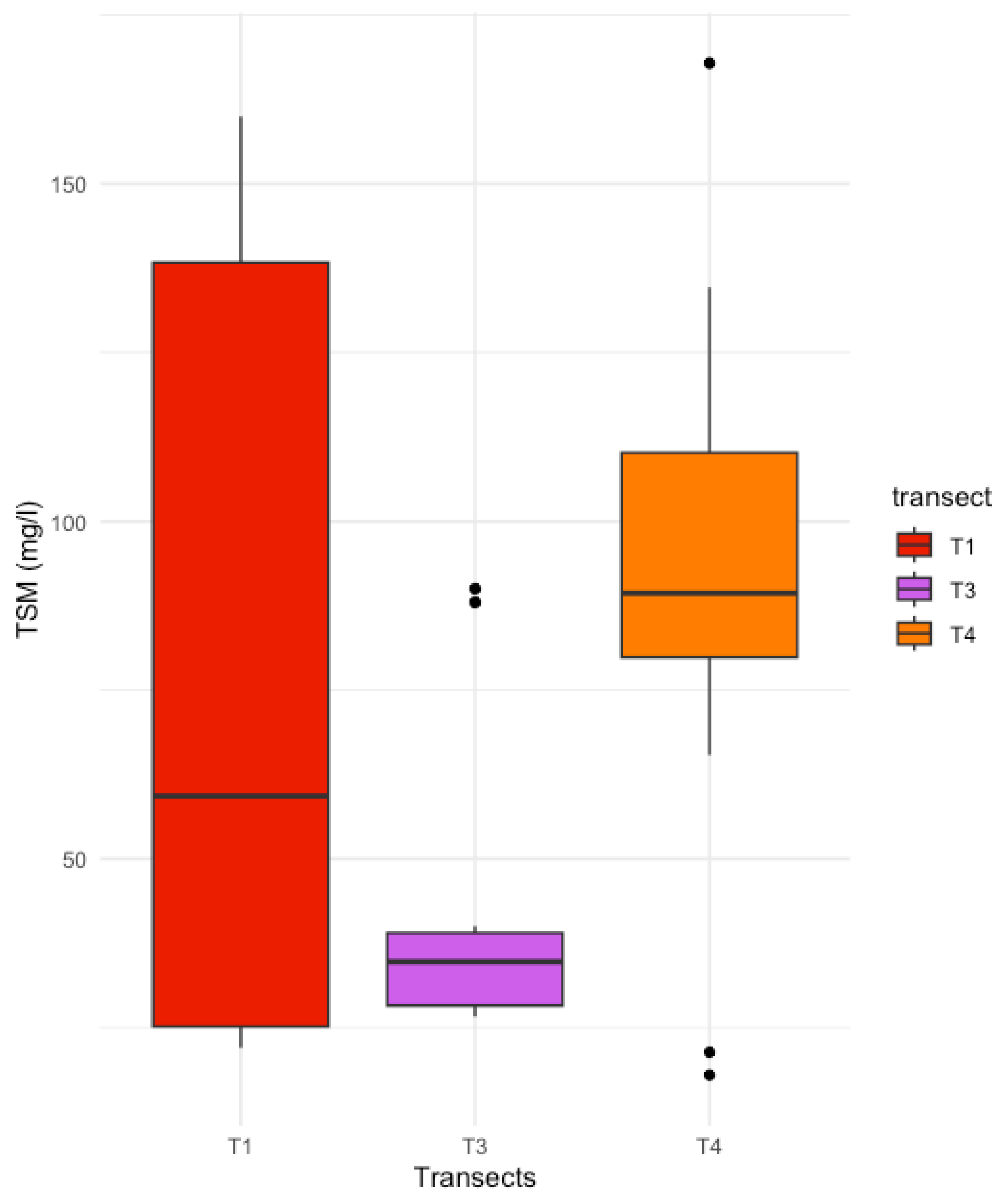
Biological Indicators
Benthic Fauna

1. Meiofauna:
Quantification of small benthic organisms (e.g., nematodes, copepods).

2. Macrofauna:
Community analysis based on abundance, Shannon Index, and Pielou Index.

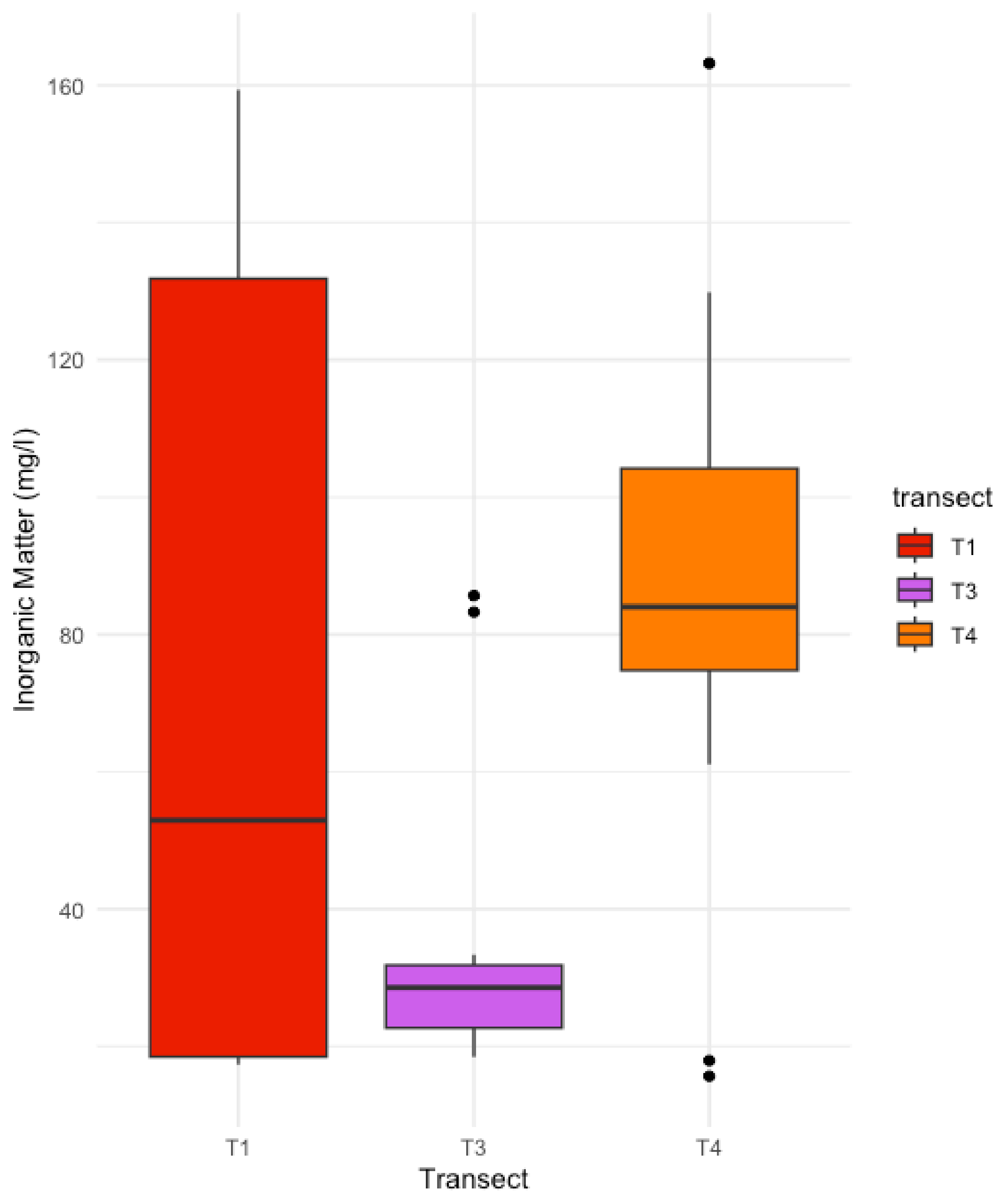
Water column

Total Suspended Matter



Tuckey post hoc
T4-T1 p-value 0.80
T4-T3 p-value 0.03 *

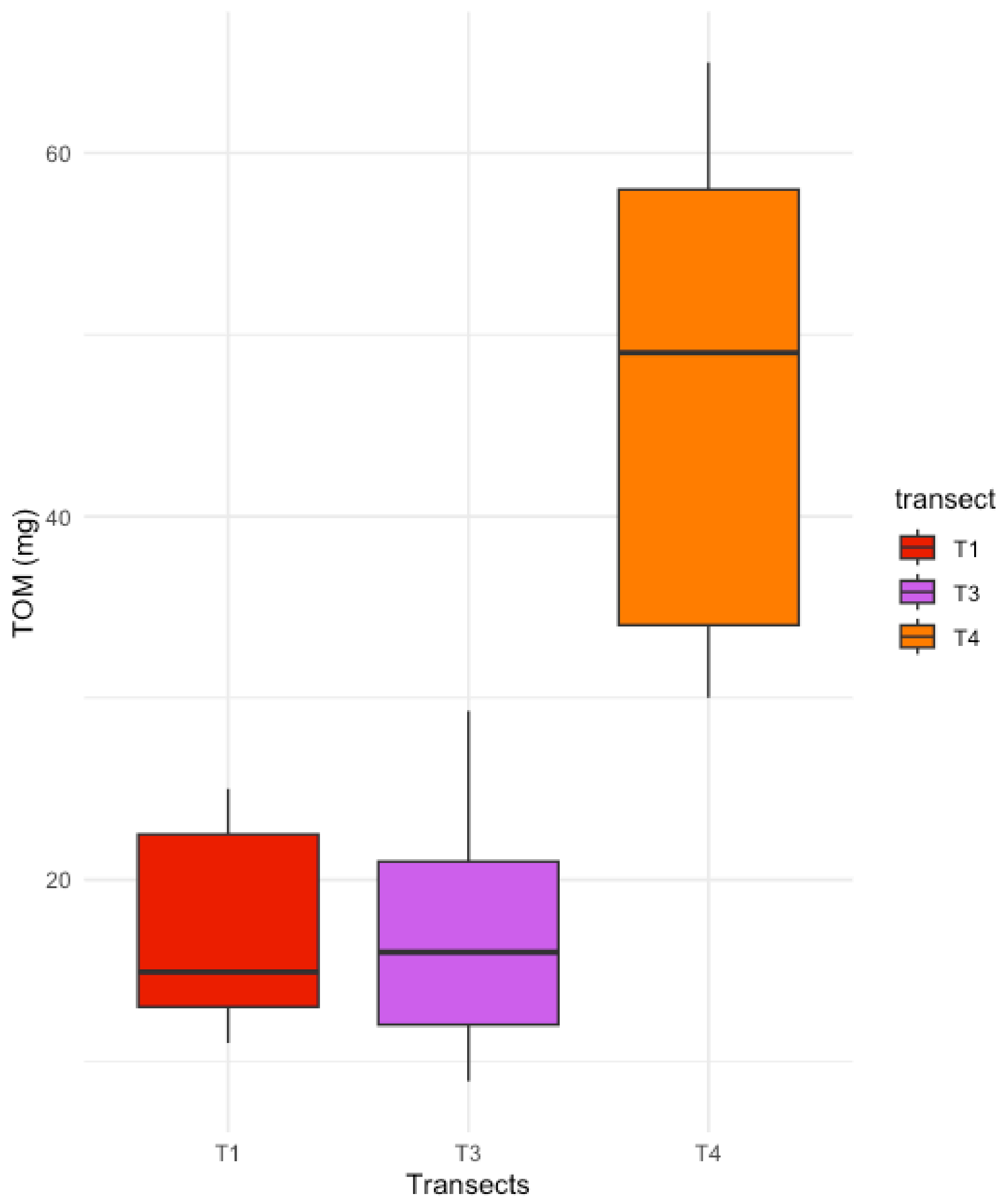
Inorganic Matter



Tuckey post hoc
T4-T1 p-value 0.78
T4-T3 p-value 0.03 *

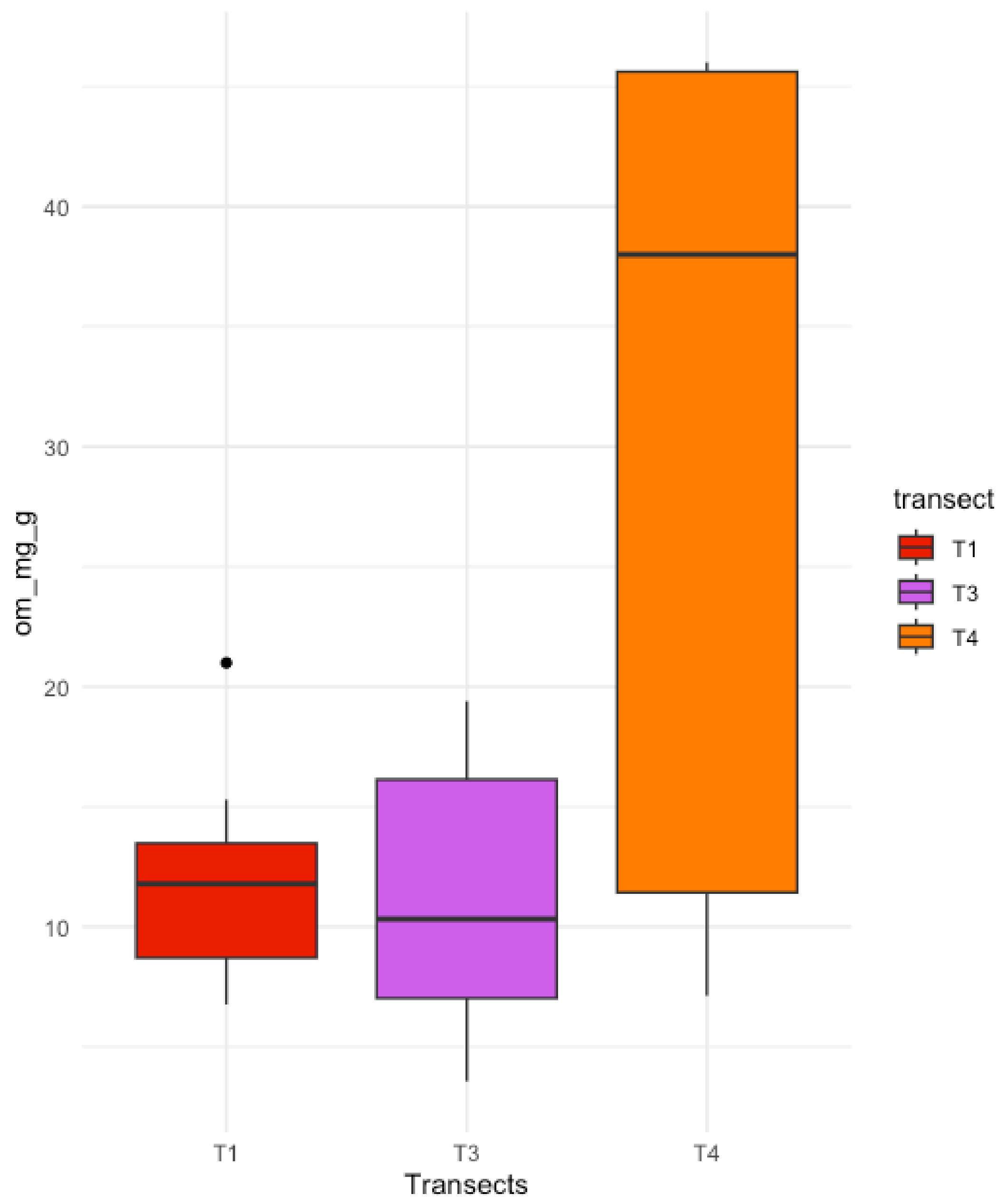
Sediment

Total Organic Matter



Tuckey post hoc
T4-T1 p-value 0,0000005 *******
T4-T3 p-value 0,0000004 *******

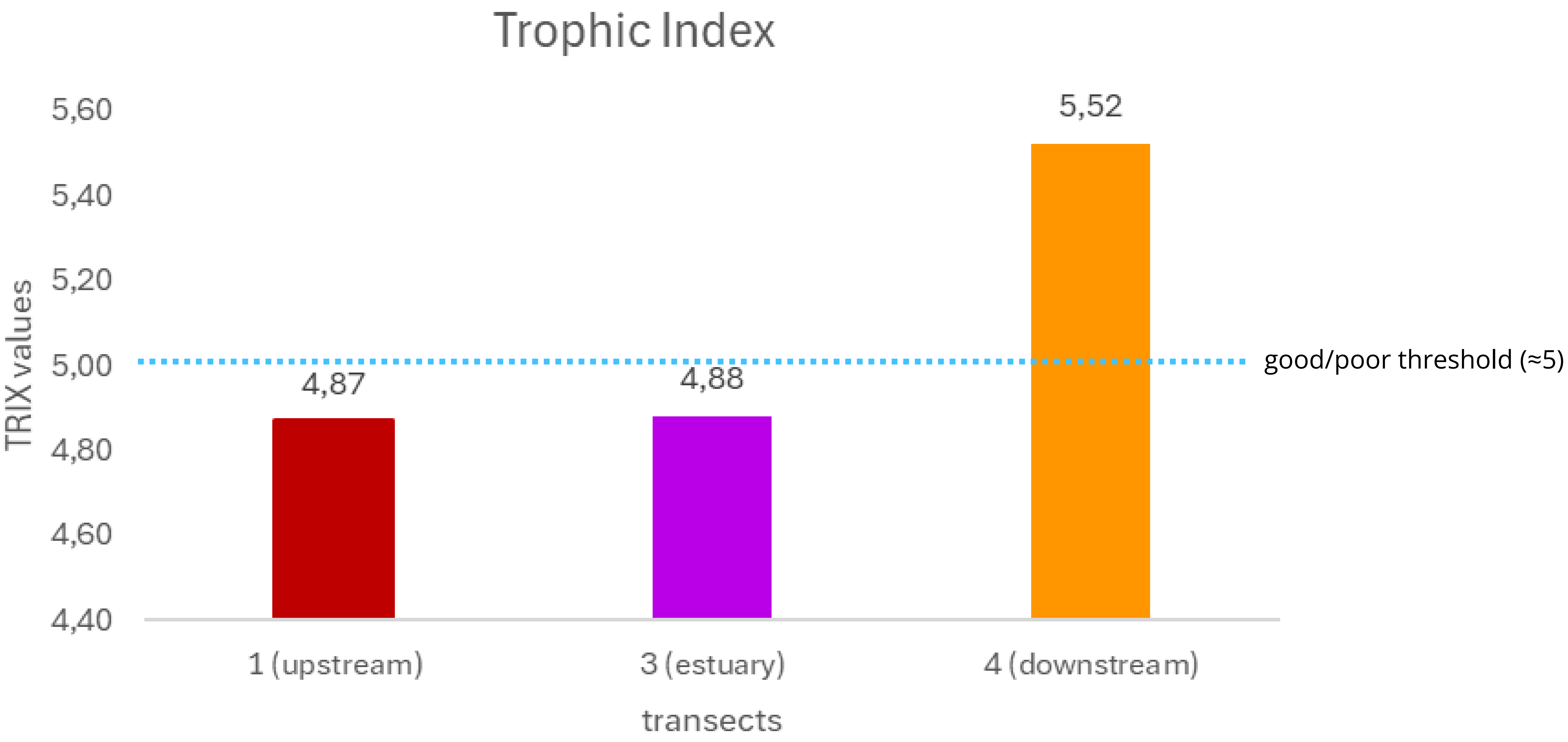
Organic Matter



Tuckey post hoc
T4-T1 p-value 0,0053 ******
T4-T3 p-value 0,0041 ******

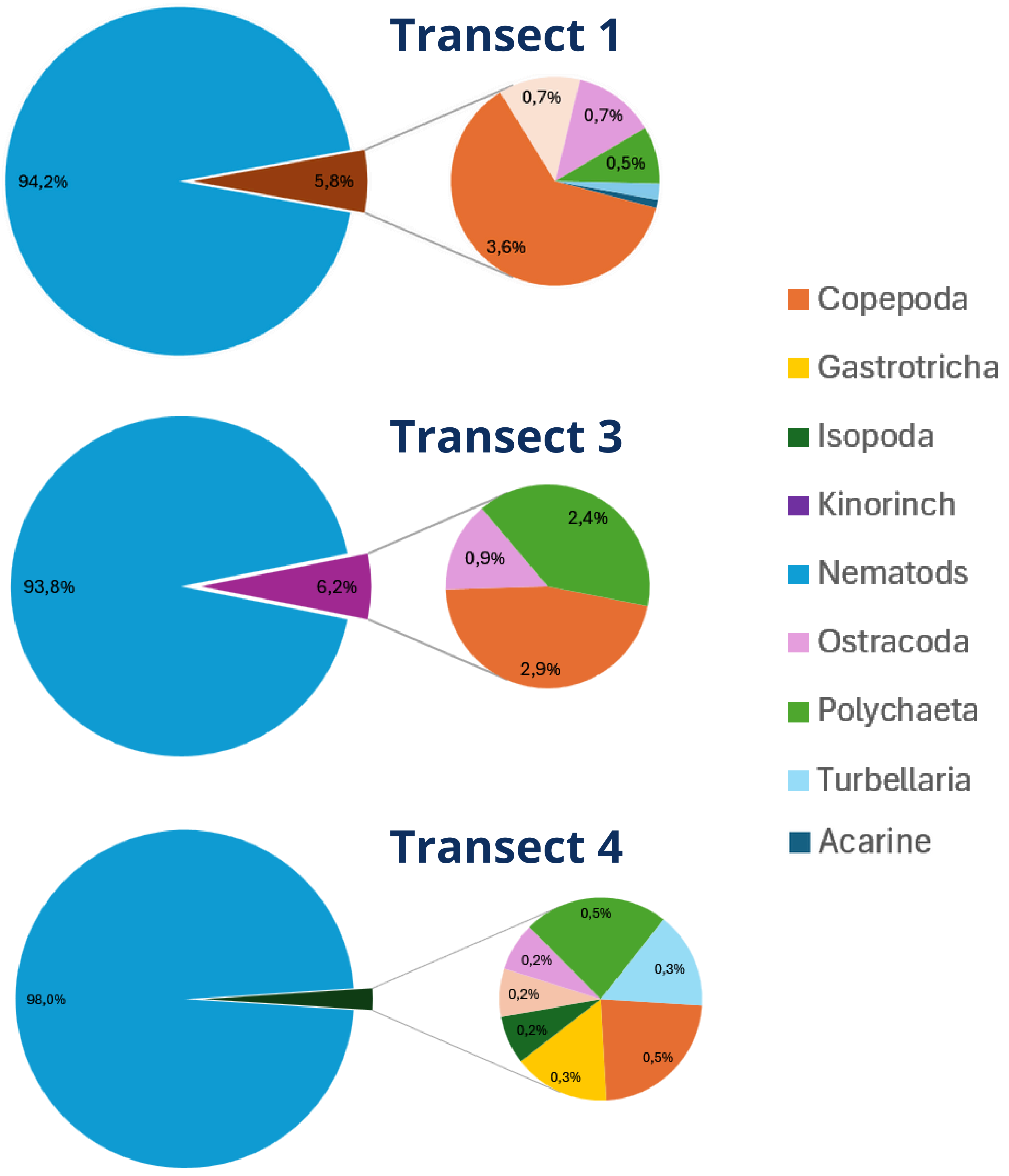
Trophic Index - TRIX

TRIX = [Log10 (Chl-a x aDO x N x P) +1.5]/1.2

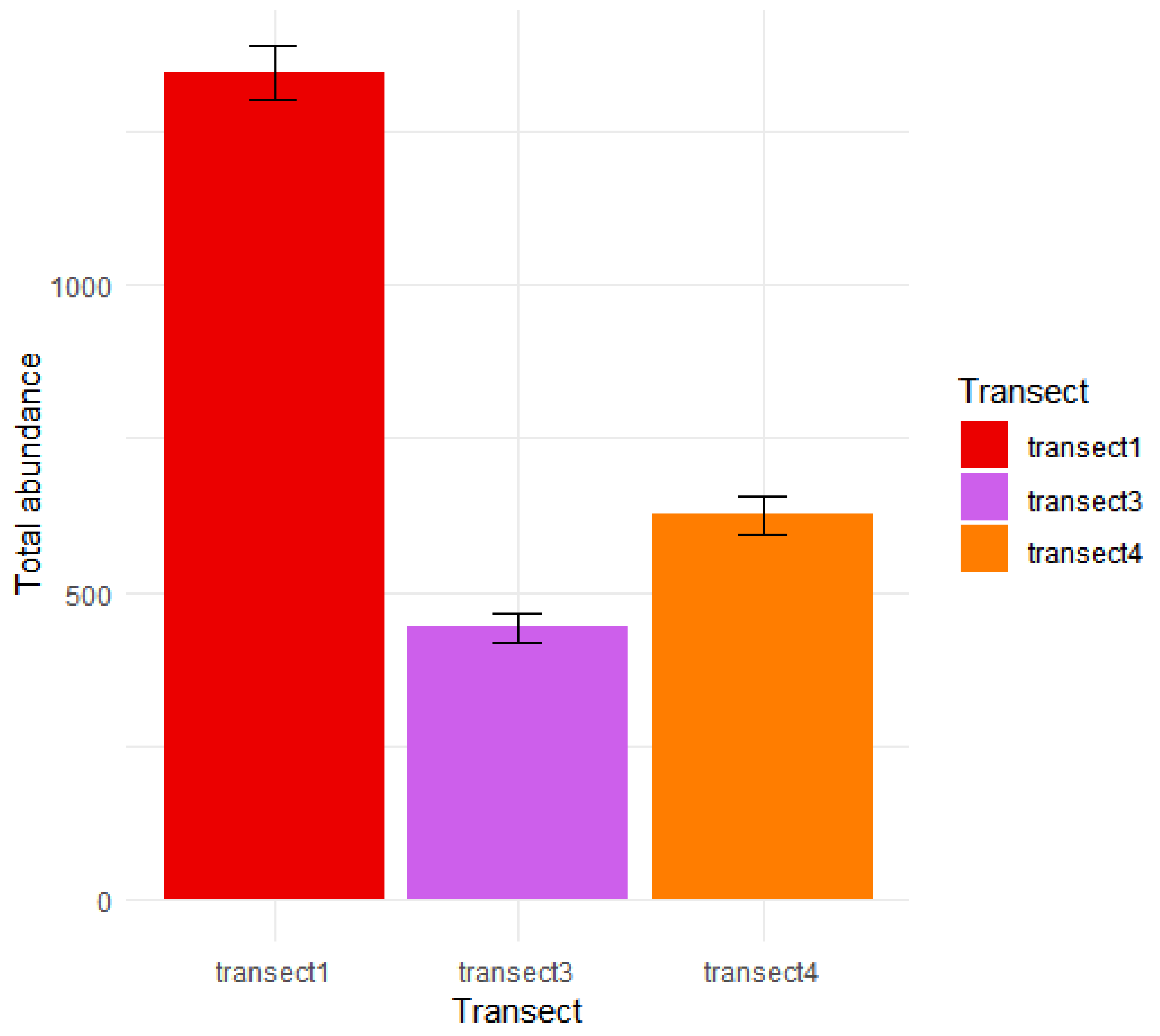


Meiofauna

Taxa distribution



Total abundance of meiofauna



Meiofauna Taxa Richness

Transect 1 = 7 taxa

Transect 3 = 4 taxa

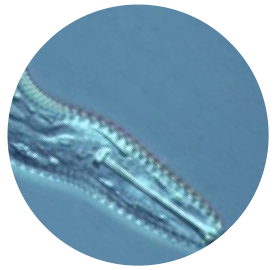
Transect 4 = 8 taxa

Non-polluted systems: 10 - 18 taxa

Polluted systems: 4 - 8 taxa

Classification proposal
<4 taxa $\Rightarrow$ bad conditions
4-8 taxa $\Rightarrow$ altered condition
8-12 taxa $\Rightarrow$ sufficient conditions
12-16 taxa $\Rightarrow$ good conditions
$>$ 16 taxa $\Rightarrow$ very good conditions

We can see an "**altered condition**" in all the analyzed transects but we don't see an increased and asymmetrical impact due to coastal currents.
We have to take in mind that often Taxa Richness is not sufficient as pollution/impact indicator.



Nematodes/Copepodes Index

It is based on the greater tolerance of Nematodes to high organic loads and consequent reduced oxygen content with Copepodes which are much more sensitive to organic loads:

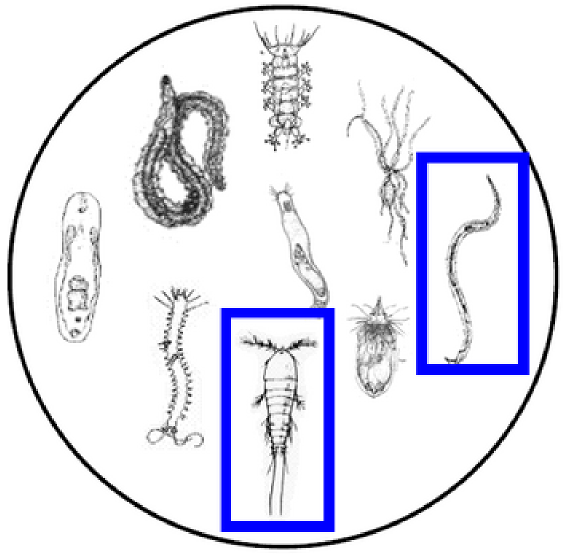
Transect 1 = N/C = 26.2

Transect 3 = N/C = 32.3

Transect 4 = N/C = 196.0

Unimpacted environmental conditions

Nematodes / Copepods ≤ 10



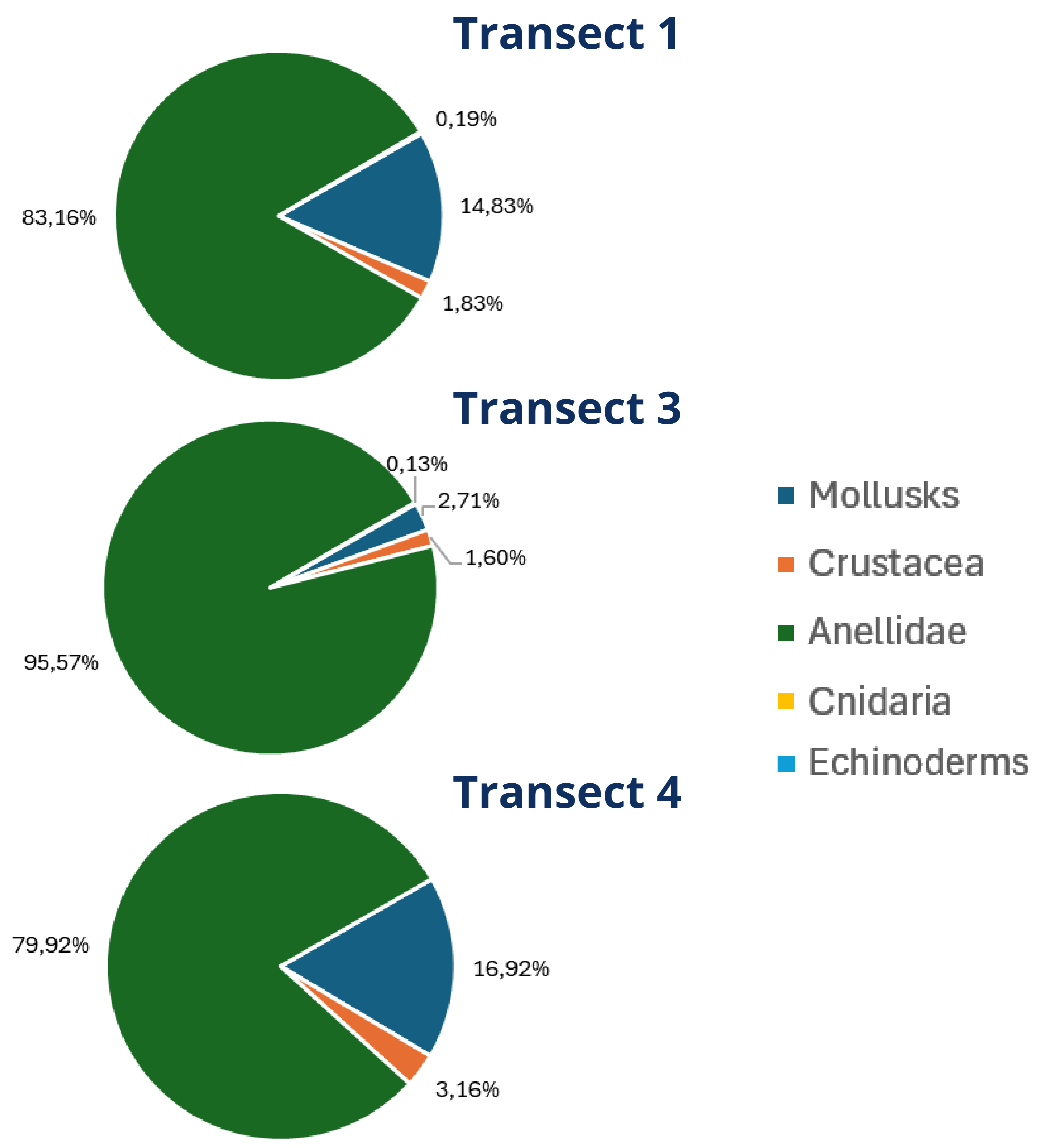
Environments characterized by organic enrichment:

20 ≤ Nematodes / Copepods ≤ 100

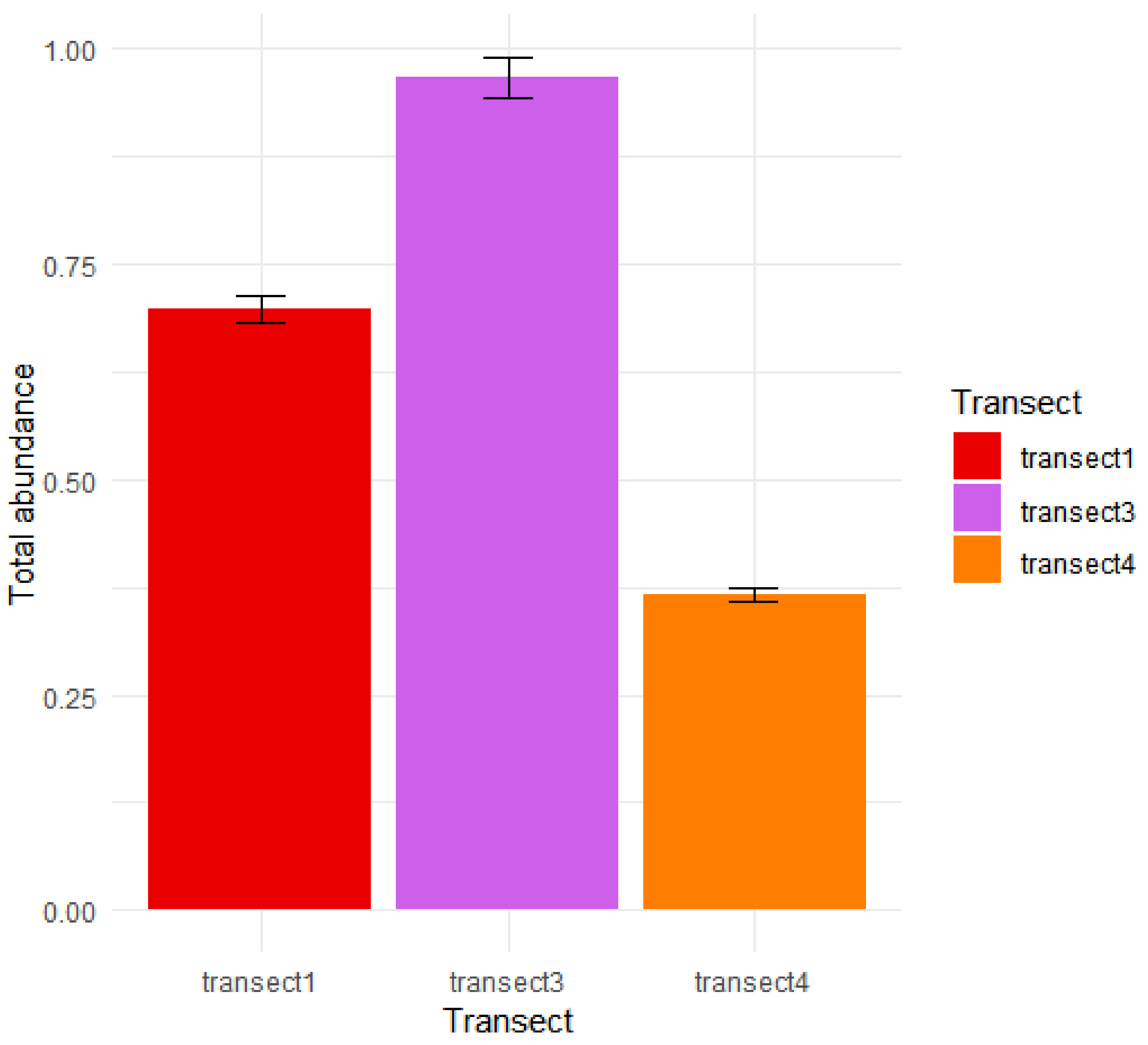
In all the transects we can observe a **N/C typical of Environments characterized by organic enrichment**, especially we can observe a C/N ratio much higher in the Tr.4 than in Tr.1 and Tr.3, which could be a hint of greater current-driven organic load in that area.

Macrofauna

Taxa distribution



Total abundance of macrofauna



Macrofauna Taxa Richness

Transect 1 = 5 taxa

Transect 3 = 5 taxa

Transect 4 = 4 taxa

Non-polluted systems: 10 - 18 taxa

Polluted systems: 4 - 8 taxa

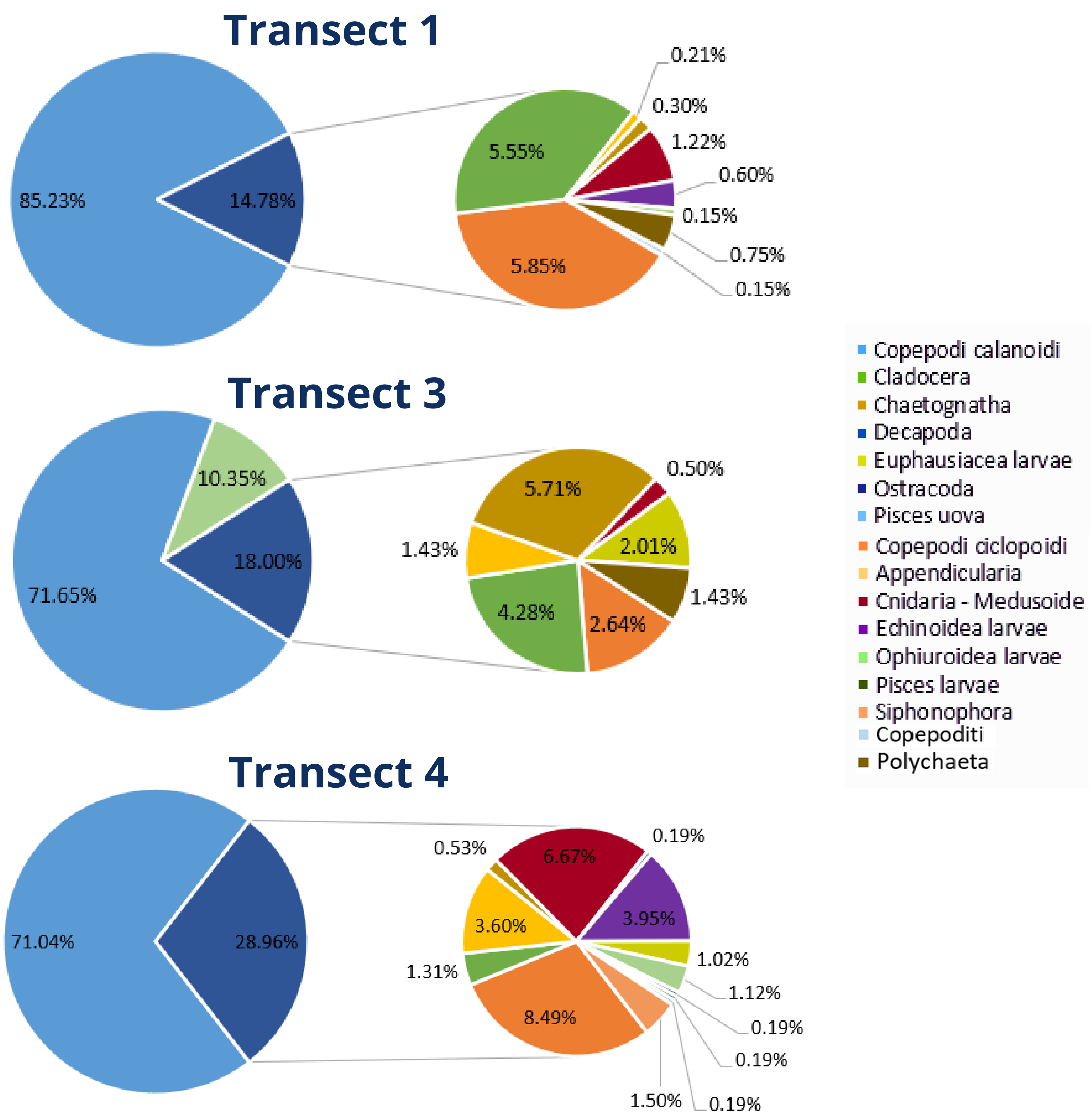
Classification proposal
<4 taxa ⇒ bad conditions
4-8 taxa ⇒ altered condition
8-12 taxa ⇒ sufficient conditions
12-16 taxa ⇒ good conditions
> 16 taxa ⇒ very good conditions

We can see an “**altered condition**” in all the analyzed transects, with a very slight reduction of taxa in transect 4.

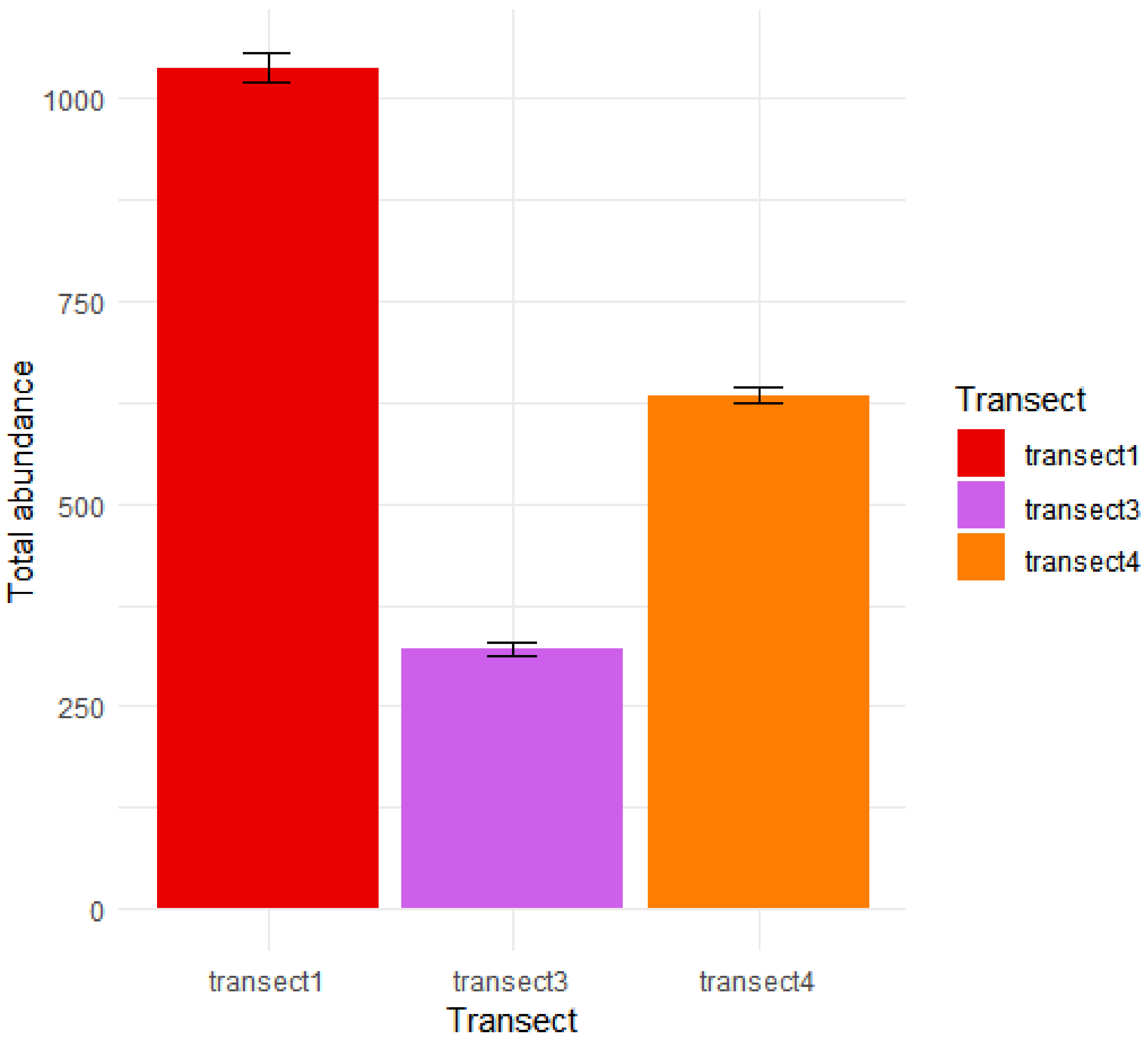
We have to take in mind that often Taxa Richness is not sufficient as pollution/impact indicator.

Zooplankton

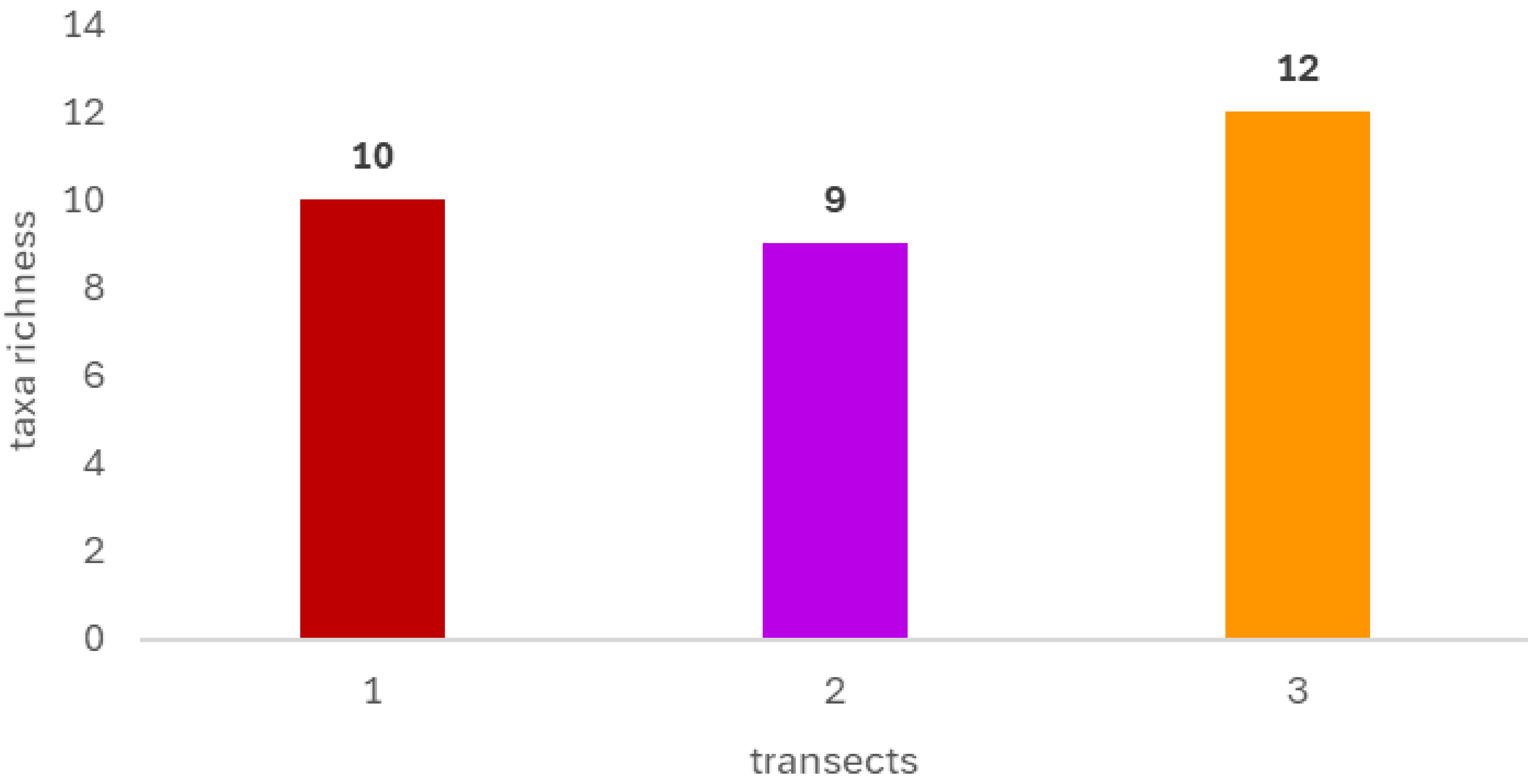
Taxa distribution



Total abundance of zooplankton

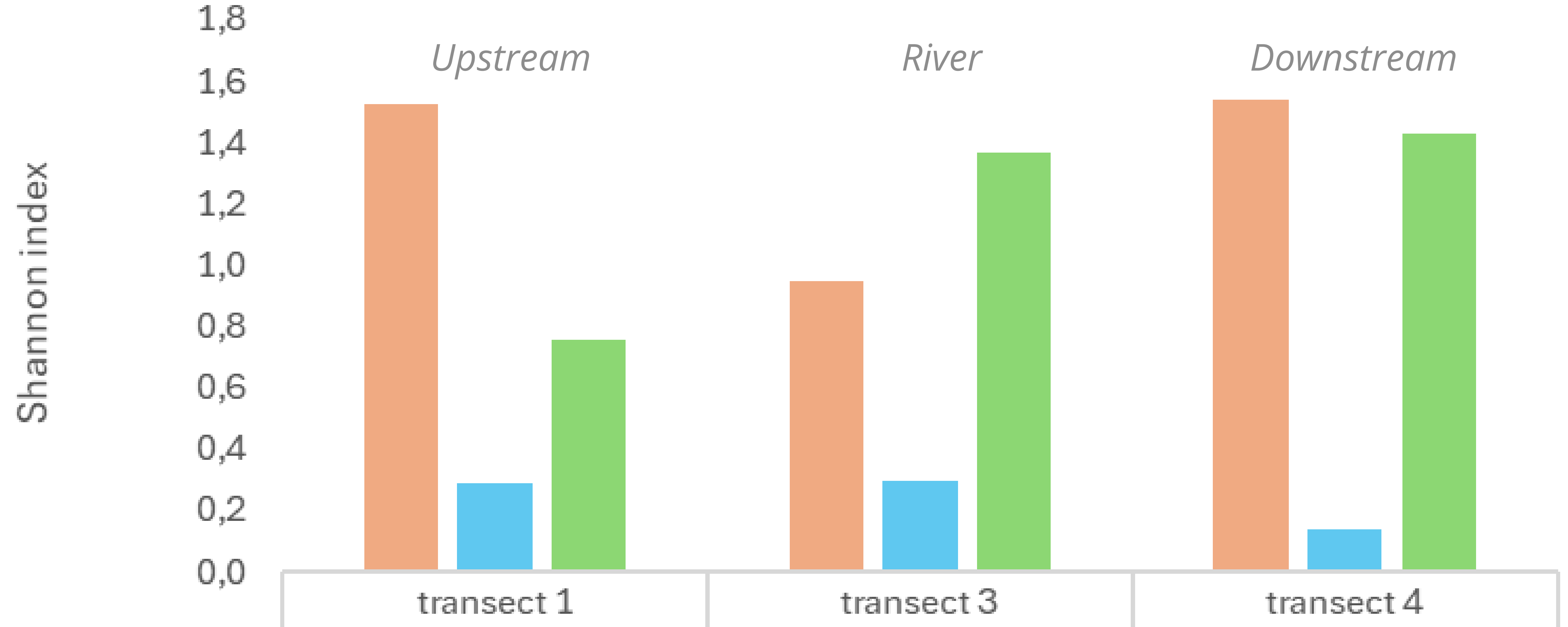


Zooplankton Taxa Richness



Zooplankton is very dynamic, sensitive to temporal, spatial, biotic and abiotic factors, so there are no universal, standardised reference thresholds for taxa richness as there are for benthic macrofauna and meiofauna.

Shannon-Wiener Index



	transect 1	transect 3	transect 4
Macrofauna	1,523	0,947	1,535
meiofauna	0,289	0,295	0,137
zooplankton	0,753	1,367	1,425

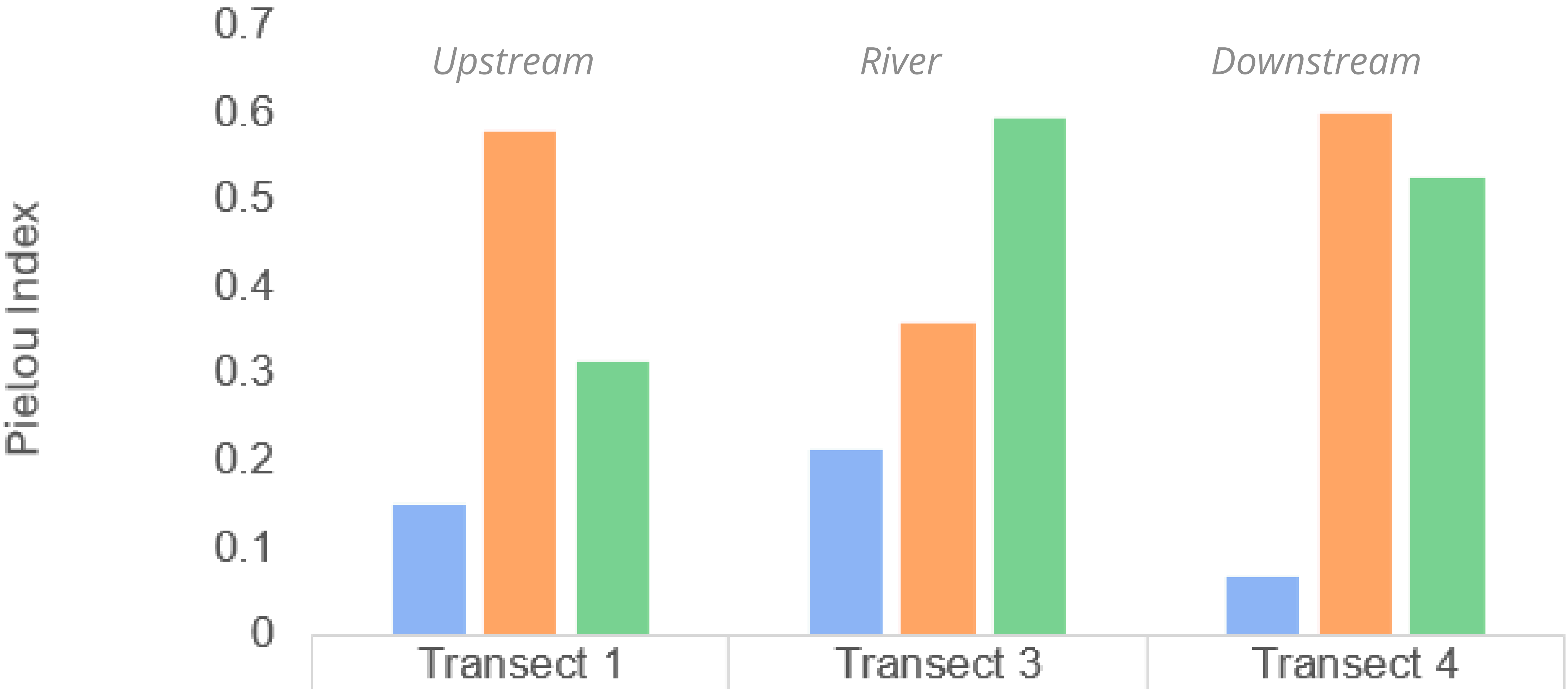
Macrofauna meiofauna zooplankton

- **MACROFAUNA: <u>Tr. 3</u>**
  → lowest macrofaunal diversity.
  → possible stress/disturbance.

- **MEIOFAUNA**
  → low overall diversity,
  nematode dominance.

- **ZOOPLANKTON: <u>Tr. 3 - 4</u>**
  → higher zooplankton diversity
  → possible link with increased phytoplankton due to nutrient enrichment.

Pielou's Evenness Index



	Transect 1	Transect 3	Transect 4
Meiofauna	0.149	0.213	0.066
Macrofauna	0.577	0.359	0.598
Zooplankton	0.314	0.594	0.526

Meiofauna Macrofauna Zooplankton

- **MEIOFAUNA**
  → low evenness across all transects but lowest at Tr. 4.
  → short-term impact index.
- **MACROFAUNA**
  → lowest evenness at Tr. 3 (estuary).
  → long-term impact index.
- **ZOOPLANKTON**
  → same reasoning as Shannon index analysis.
- Overall the analysis, expecially close to the river (Tr. 3), reports low Pielou's index that represent an ecologic disequilibrium.

To sum up

ANALYSIS	WATER	SEDIMENT
TRIX	✓	–
TSM	✗	–
IM	✗	–
TOM	–	✓
OM	✗	✓

To sum up

ANALYSIS	MEIOFAUNA	MACROFAUNA	ZOOPLANKTON
Total abundance	✓	✓	✓
Taxa richness	✗	✗	—
Ne/Co	✓	—	—
Shannon-Wiener Index	✓	✓	✓
Pielou’s Evenness Index	✓	✗	✓

Hypotheses

Ho

No significant impact of river runoff on environment or fauna; no current-driven pattern.

H1

River runoff affects environmental quality and fauna, with stronger, asymmetrical impacts downstream due to coastal currents.

THANK YOU!

Bibliography and Sitography

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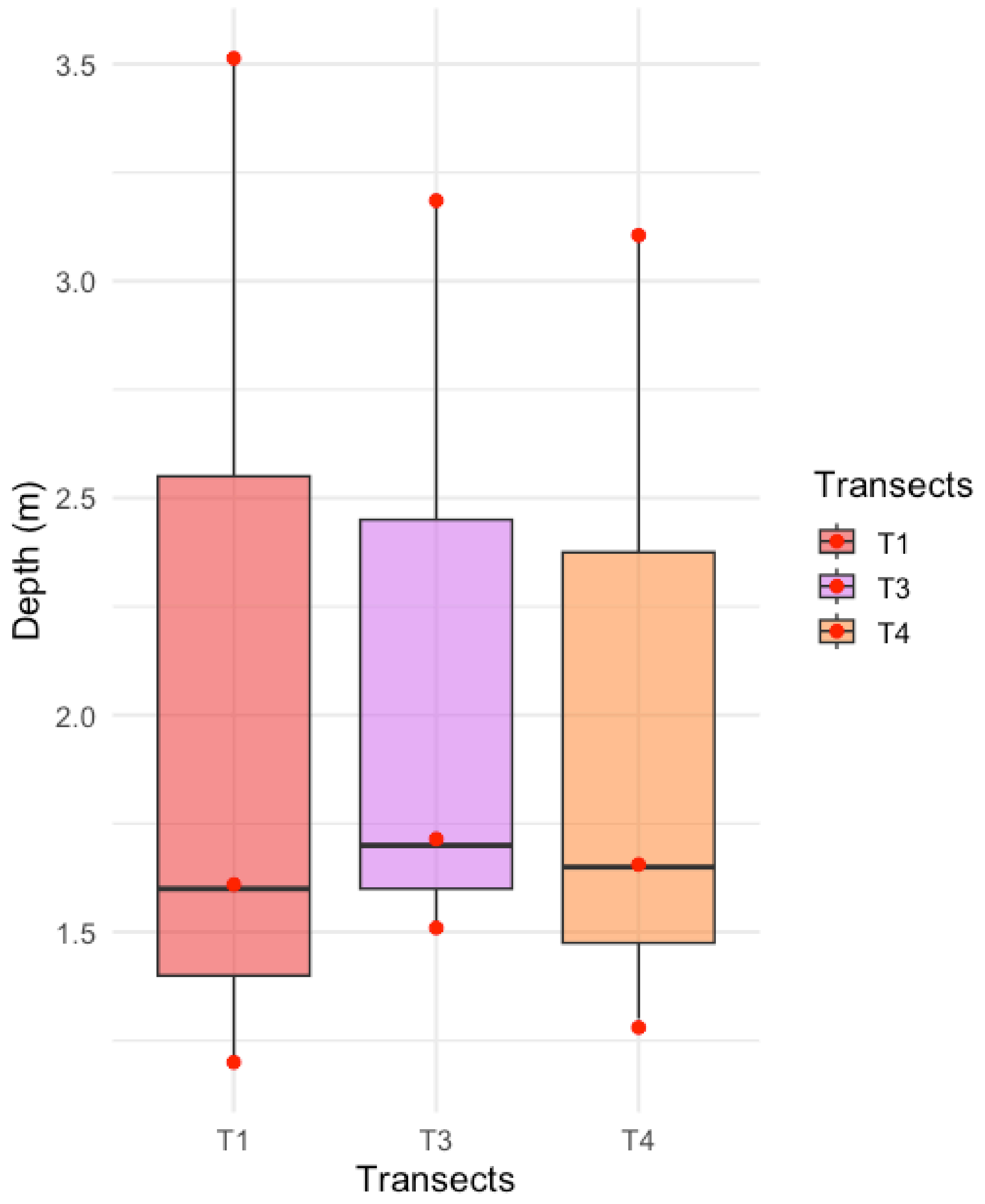
- https://www.arpa.marche.it/indicatori-ambientali?id=1046

Secchi Disk



F value molto basso (0.01 → differenze minime rispetto alla variabilità interna ai gruppi.)
P-value = 0.99 no differenza significativa.
Tukey post-hoc no diffreenze significativetra coppie.